PREDICTING OF 'DEKAN' STATUS AMONG GRADUATES USING LOGISTIC REGRESSION

MDA3035 | SEM II 24/25 | GROUP 1

NUR SYUHADAH BINTI ZAIMI S74290
NUR AZLINA NATASHA BINTI RUSLY S74322
DANISH HATEEM BIN JUMAT S74742
PRENISHA A/P SIVADEV S76045

01 INTRODUCTION

Graduation rates are a critical measure of educational success,this study aims to predict whether a university student will graduate with "dekan" status based on their background and entry information. Using machine learning, we identify which factors such as entry CGPA, income group, and admission stream are most associated with high academic performance. Logistic regression is well-suited for this binary classification task, offering clear insights into which factors significantly influence graduation outcomes. The goal is to leverage these findings to support data-driven interventions that can enhance student success and reduce dropout rates.

PROBLEM STATEMENT

Many students do not graduate with dekan status. Thus, there is a need to develop a predictive model using student entry data to identify key factors influencing academic excellence.

02 DATASET DESCRIPTION

The dataset contains information about university graduates, including their gender, CGPA when they entered university, CGPA at graduation, the type of entry stream they came from (like STPM, Diploma, or Matric), and their household income group (B40, M40, or T20). It also shows their academic status, such as whether they graduated or were still active.

CREATION OF TARGET VARIABLE

- DEKAN = 1 IF CGPA >= 3.5
- BUKAN DEKAN = 0 IF CGPA < 3.5

Only students who successfully graduated were selected for analysis.

JANTINA	STATUS	CGPA KEMASUKAN	ALIRAN KEMASUKAN	KATEGORI PENDAPATAN	CGPA BERGRADUAT	DEKAN
LELAKI	BERGRADUAT	2.92	ALIRAN SASTERA STPM	B40	3.57	1
LELAKI	BERGRADUAT	3.52	ALIRAN SASTERA STPM	M40	3.55	1
LELAKI	BERGRADUAT	2.94	ALIRAN SASTERA STPM	B40	3.01	0
PEREMPUAN	BERGRADUAT	3.52	ALIRAN SASTERA STPM	B40	3.62	1
PEREMPUAN	BERGRADUAT	3.74	ALIRAN SASTERA STPM	B40	3.76	1

03 METHODOLOGY

1 Data Preprocessing

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()

model.fit(x_train_scaled, y_train.ravel())
```

```
print('Model intercept: ', model.intercept_[0])
print('Model coef: ', model.coef_[0])
```

```
Model intercept:  0.036732388375626 52
Model coef:  [0.50394509 0.13616868 0.06105582 0.04159838]
```

The first line prints the intercept, which is the model's base prediction when all input values are zero. The second line prints the coefficients, which show how much each feature (like CGPA or gender) affects the chance of being predicted as "Dekan". A higher positive value means the feature increases the chances, while a negative value decreases it.

2 Feature Selection

```
'CGPA KEMASUKAN',
'jantina_encoded',
'pendapatan_encoded',
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'DEKAN'.
```

The variable x is assigned a subset of columns from the dataset that will be used as features: 'CGPA KEMASUKAN', 'jantina_encoded', 'pendapatan_encoded', and 'aliran_encoded'. The variable y is set as the 'DEKAN' column, which is the target or label the model will learn to predict, indicating whether a student is on the Dean's List or not.

3 Model Training

59.89 %

Model Testing

60.47 %

4 Performance Evaluation

Testing Data

	precision	recall	f1-score	support
0	0.60	0.54	0.57	363
1	0.61	0.67	0.64	396
accuracy			0.60	759
macro avg	0.60	0.60	0.60	759
weighted avg	0.60	0.60	0.60	759

Training Data

	precision	recall	f1-score	support
0	0.60	0.57	0.58	1118
1	0.60	0.63	0.61	1156
accuracy			0.60	2274
macro avg	0.60	0.60	0.60	2274
weighted avg	0.60	0.60	0.60	2274

The report includes key metrics for each class:

- precision – how many predicted positives are actually correct
- recall – how many actual positives were correctly predicted
- f1-score – the harmonic mean of precision and recall
- support – the number of true instances for each class

04 RESULTS

FIGURE 1

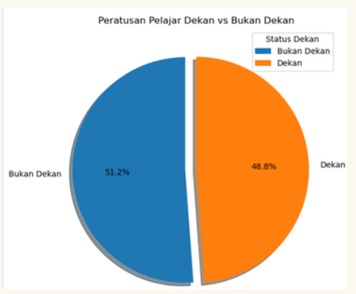


FIGURE 2

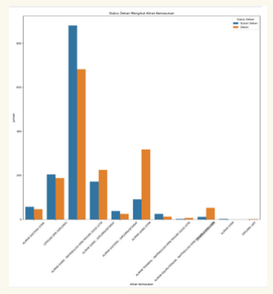


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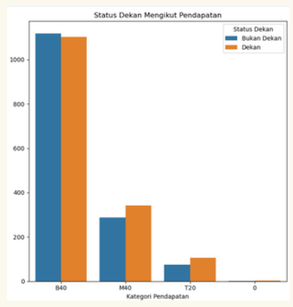


FIGURE 4

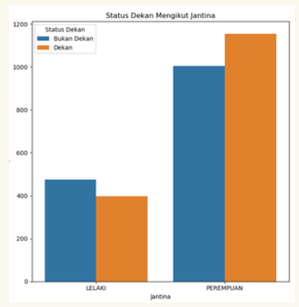
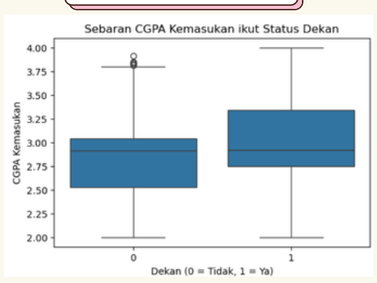
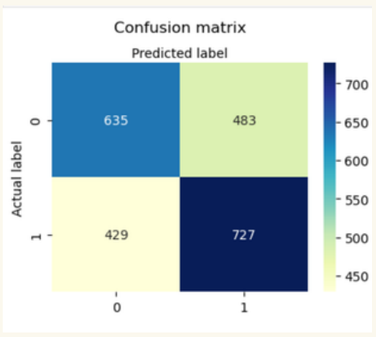


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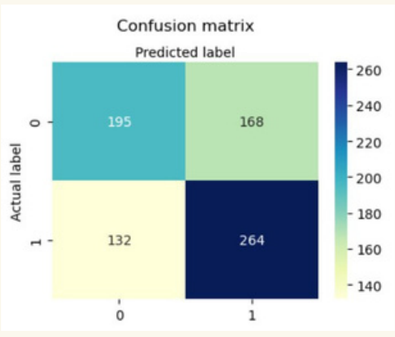


TRAIN DATA



The training data makes up 80% of the full dataset and is used to help the model learn the relationship between student background factors and their likelihood of getting "DEKAN" status. During training, the model achieved around 59.9% accuracy. It was slightly better at recognizing students who did not make the "DEKAN" , which suggests the model may have learned more patterns from that group.

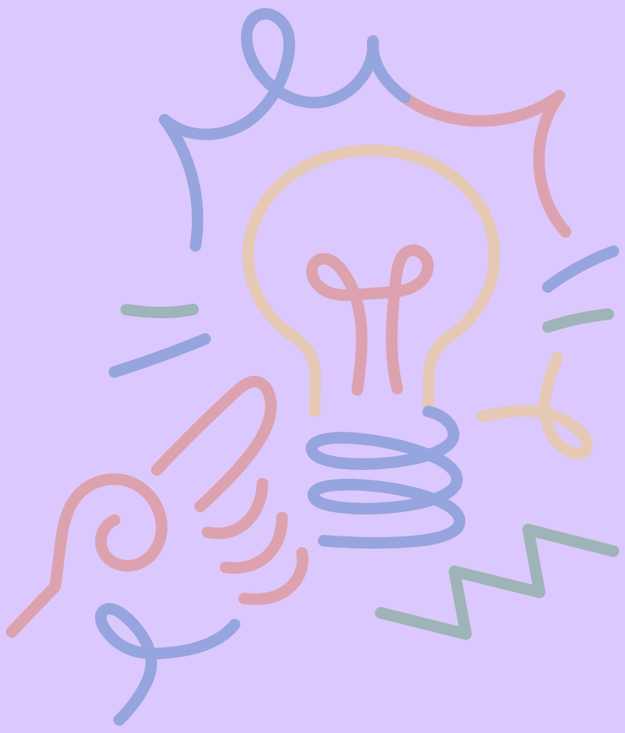
TEST DATA



The testing data represents the remaining 20% of the dataset and is used to check how well the model performs on new, unseen cases. The model reached about 60.5% accuracy on this test set. The confusion matrix show that the model made some mistakes in both directions which sometimes predicting a student would get "DEKAN" when they didn't, and vice versa.

05 CONCLUSION

The logistic regression model successfully predicts Chancellor's Award eligibility (CGPA $\geqslant$ 3.5), with entry CGPA as the primary determinant. Additional factors such as income category and academic stream further influence outcomes, highlighting opportunities for targeted academic interventions. This data-driven approach demonstrates the value of predictive analytics in enhancing student success strategies.



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GROUP 1

MDA3053 INTRODUCTION TO MACHINE LEARNING

SEM II 24/25

LOGISTIC REGRESSION



MEMBERS

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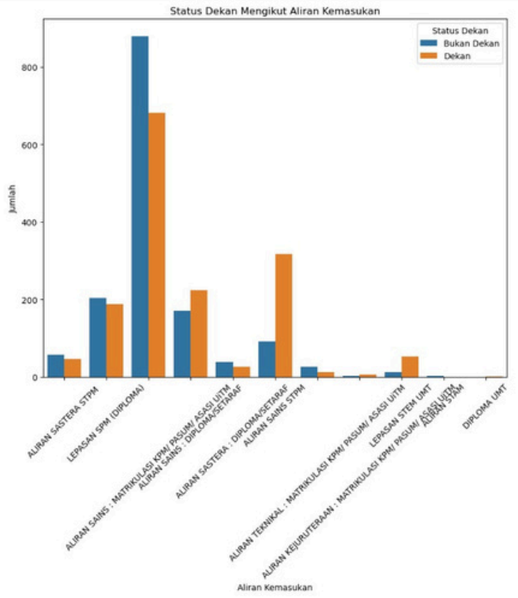


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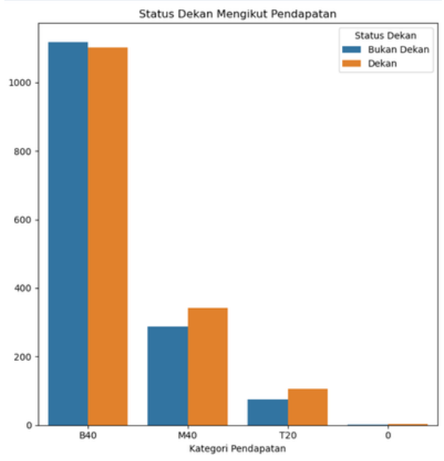


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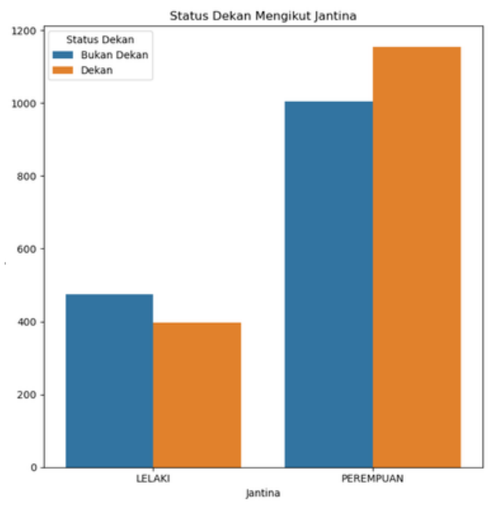
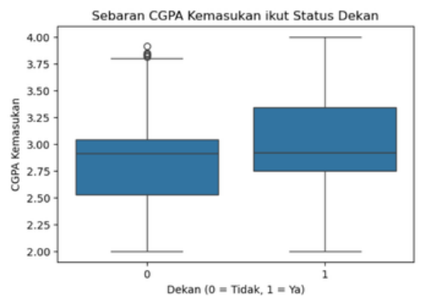
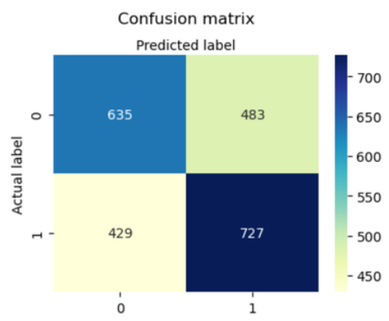


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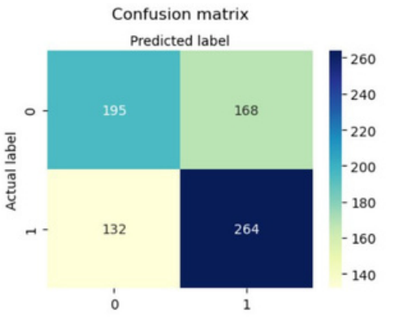
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CREDITS: DR SHALELA